

SIDDHARTH P / AI/ML ENGINEERING STUDENT & LOCAL-FIRST AI SYSTEMS BUILDER

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POSITIONING

Final-year B.E. AIML student at BNMIT focused on local-first AI systems, RAG/document intelligence, voice AI, computer vision, and privacy-conscious product engineering.

EDUCATION

B.E. Artificial Intelligence & Machine Learning
BNM Institute of Technology, Bengaluru
2023-2027 | CGPA: 8.61/10
Pre-University: PCMC, Sri Chaithanya PU College
ICSE: Sri Vani Education Centre

CORE STACK

Python, TypeScript, JavaScript
FastAPI, Flask, Next.js, React
SQLite, Chroma, BM25, RRF
Ollama, local LLMs, faster-whisper
OpenCV, MediaPipe, scikit-learn
PyTorch, TensorFlow, Grad-CAM
Firebase, Docker, Git/GitHub
Flutter, Algorand, IPFS

WHAT I BUILD

Offline RAG & study workspaces
Voice assistants & OS automation
Computer vision safety systems
Medical imaging ML prototypes
AI-assisted learning/devtools
Privacy-first data workflows

PROFESSIONAL SUMMARY

AI/ML engineering student and systems builder with a portfolio spanning offline academic intelligence, local voice automation, multimodal stress detection, rider-safety edge AI, medical imaging, decentralized MRV, and data-quality automation. Strongest fit: AI engineering / applied ML / AI product engineering roles where product thinking, backend architecture, retrieval, automation, and practical model evaluation matter more than generic CRUD development.

EXPERIENCE

AI/ML Intern - Equinox Pvt Ltd | June 2025 - July 2025

- Worked in a human-lab research setting on AI-based facial analysis methods for stress detection in pilots and prolonged aviation-stress scenarios.
- Explored facial-analysis signals and stress-related patterns, linking research objectives to practical non-invasive monitoring workflows.

FEATURED TECHNICAL PROJECTS

NirmiqResearchOS | Offline-first Academic Intelligence System [GitHub](#)

Python | FastAPI | Next.js | SQLite | Chroma | BM25 | Ollama | RRF

- Built a local academic intelligence workspace using FastAPI, SQLite, hybrid BM25 + Chroma vector retrieval, Reciprocal Rank Fusion, reranking, and Ollama-backed local generation.
- Implemented grounded document workflows: ingestion, chunk indexing, PDF/text/Markdown/image upload, citation jump links, evidence trails, document drilldown, query comparison, and retrieval profiles.
- Designed 10 study modes including research, deep research, research paper, exam answer, revision notes, important questions, compare concepts, and study guide.

NirmiqEcho | Local-first Windows Voice Assistant [GitHub](#)

Python | faster-whisper | VAD | Ollama | Windows Automation | TTS

- Created an offline JARVIS-style assistant for Windows with faster-whisper STT, system-tray control, F9 listening toggle, app/file search, WhatsApp messaging, math, timers, screenshots, volume, and system controls.
- Implemented a two-layer understanding stack: 90+ offline command patterns plus optional local Ollama fallback that maps free-form speech into whitelisted commands.
- Added privacy and safety hardening: on-device STT/TTS, GPU/CPU fallback ladder, self-healing mic checks, no shell=True/eval, allow-listed launches, and spoken confirmation for destructive actions.

SmartHelm | AI Rider Safety & Fleet Monitoring [GitHub](#)

Python/Flask | Kotlin | MediaPipe | Firebase | Raspberry Pi | MAX30102 | Firestore

- Developed an edge-first drowsiness detection system for motorcycle delivery riders using MediaPipe FaceLandmarker, Eye Aspect Ratio, PERCLOS tracking, on-device alerts, and fleet telemetry.
- Architected Raspberry Pi helmet backend, Android foreground detection service, Firebase Firestore publisher, Cloud Function SMS routing, and real-time browser fleet dashboard.
- Designed privacy-preserving telemetry: no raw video/full-face frames leave the device; only eye state, PERCLOS, heart-rate/SpO2, eye-region strip, GPS, and alert metadata are shared.

EVIDENCE SIGNALS

13 public GitHub repositories
12+ projects shipped/building
1 AI/ML internship
Local-first systems portfolio
Public portfolio + GitHub proof-of-work

SELECTED METRICS

Stress Detection: 96.57% accuracy, 96.66% F1, 99.39% ROC-AUC
Brain Tumour MRI: 98.95% accuracy/F1, 99.92% ROC-AUC in recorded 5-fold CV run
NirmiqEcho: 90+ offline command patterns
SmartHelm: 60s rolling PERCLOS tracker

RESEARCH / ACADEMIC

IEEE-format BNMIT review paper on multimodal stress detection, rPPG/HRV-style signals, facial analysis, and non-invasive stress inference.
Final-year project contribution: literature survey, requirement specifications, architecture design, Gantt planning, and conclusion ownership.

ROLE FIT

Applied AI Engineer Intern
AI/ML Intern
RAG / Document AI Intern
Voice AI / Assistant Builder
Computer Vision ML Intern
AI Product Engineering Intern

WORK STYLE

Proof-of-work oriented
AI-assisted development workflow
Local/private AI focus
Product + engineering mindset
Bengaluru/remote availability

ADDITIONAL PROJECT DEPTH

Multimodal Stress Detection System v3.0 | Webcam-only stress + drowsiness ML GitHub

Python | OpenCV | MediaPipe | rPPG/POS | scikit-learn | Random Forest

- Built a privacy-first, on-device pipeline combining MediaPipe 468-landmark facial analysis, rPPG heart-rate estimation, HRV RMSSD, PERCLOS, brow furrow, head pose, blink features, and a 9-D Random Forest fusion classifier.
- Added continuous 0-100 stress scoring, label-buffer vote decay, drowsiness alerts, CSV logs, reconnect logic, HTML reports, 66 tests, and reported 96.57% accuracy / 96.66% F1 / 99.39% ROC-AUC.

Automated Brain Tumour Detection | Hybrid CNN + Vision Transformer MRI classification GitHub

PyTorch | ResNet50 | Vision Transformer | Radiomics | Grad-CAM | Gradio

- Implemented a research/education MRI classifier for glioma, meningioma, no tumour, and pituitary classes using ResNet50 features, patch embedding, 6-layer ViT, radiomics branch, fusion classifier, and Grad-CAM explanations.
- Includes config-driven training, optional SSL hooks, augmentation, local Gradio UI, Hugging Face model-card alignment, CV scripts, and recorded 98.95% accuracy/F1 with 99.92% ROC-AUC in a 5-fold run.

NirmiqLearn OS | AI-assisted developer learning OS GitHub

Next.js 16 | TypeScript | Tailwind 4 | SQLite | Drizzle | Zod | Zustand | MCP SDK

- Built a local-first IDE companion using MCP integrations with Claude Code, Cursor, and Windsurf to convert AI-assisted coding into debug logs, explain-back questions, DSA/CS concept links, learning maps, workspaces, and Markdown exports.
- Uses local SQLite storage, localhost-only app binding, stdio MCP server, no telemetry, BYOK AI tools, and 10 MCP tools.

Project Aarna | Decentralized blue-carbon MRV on Algorand GitHub

Algorand | AlgoPy | Puya | React | TypeScript | Vite | Tailwind | IPFS

- Built a D-MRV platform where developers submit IPFS evidence, validators approve/reject on-chain, AARNA Algorand Standard Assets are issued as carbon credits, and credits can be listed/traded through a marketplace.
- Implemented an ARC-4 AlgoPy/Puya smart contract with admin/validator/developer roles, on-chain project registry, ASA creation, inner transactions, and Pera/Defly/Exodus wallet integration.

OTHER BUILDS

PayGuard DQ | Payment data quality scoring GitHub

Python | FastAPI | SQLAlchemy | SQLite | Next.js | TypeScript | Recharts

- Built a seven-agent payment-data quality system across completeness, uniqueness, validity, consistency, timeliness, integrity, and reconciliation; exports dbt/Great Expectations tests and stores only metadata/scores/aggregates.

AnonThera | Anonymous student support platform GitHub

React | Vite | Firebase | Gemini API | Firestore | Framer Motion

- Created a student-focused anonymous support platform with Gemini AI chat, mood tracking, peer matching, multilingual support, verified helplines, reporting, chat timeouts, and data export using React/Firebase.

AGION | Flutter personal ascension system GitHub

Flutter | Dart | Riverpod | Hive | CustomPainter | GitHub Actions

- Built a mobile-first gamified self-improvement app with XP ring HUD, quick XP actions, rank progression, daily streaks, offline Hive persistence, Riverpod state management, and level-up toasts.